G10 PA Resource Department

Semester 2 Finals Review Guide

Precalculus H Review Guide

Writer: Milly 10 (1)

Editor: Chris 10 (11)

Range: Chapter 9,10,13(Mostly), Chapter 7,8

Note: (Review Guide/Review practices/Practice Set) are UNOFFICIAL sources of reference and do NOT include or hint at questions in the actual exam!

Chapter 7

Measurement of Angles

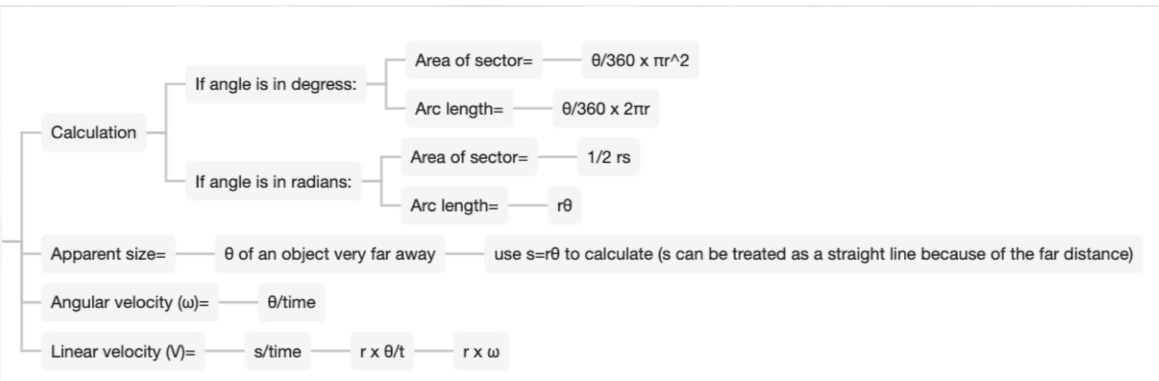
- ✓ Definition of Radian, Standard Position, Coterminal Angles
- ✓ Memorize Special Angles

Sectors of Circles

- ✓ Area of the Sector in a circle (Angle is in degrees/ Angle is in Radians)
- ✓ Apparent Size
- ✓ Angular Velocity and Linear Velocity

Trig Functions

- ✓ Definitions of Sine and Cosine Function + Unit Circle
- ✓ Reference Angles and Finding Sines and Cosines Special Angles
- ✓ Definitions and Graphs and Calculations of Sin, Cos, Tan, Cot, Sec, Csc Functions (Ex. Give you one trigonometry value and domain, find the values of the other five trigonometric functions)
- ✓ Definitions and Graphs of Inverse Trigonometric Functions



Chapter 8

Solving Trig Functions

- ✓ General Solutions
- ✓ Inclination and Slope

- ✓ Solving Trig Inequality

Shifting and Dilation

- ✓ Vertical Dilation (Stretch & Shrink)
- ✓ Horizontal Dilation (Stretch & Shrink)
  - Finding period and amplitude
- ✓ Solving trig equations (after shrink or stretch)
- ✓ Modeling Periodic Behavior
  - Transformation
  - Solving application problems
- ✓ Relationships Among the Functions
  - Reciprocal Ratio Identities, Pythagorean Relationships, Cofunction Relationships

$$\sin^2\theta + \cos^2\theta = 1 \qquad \cos^2\theta = 1 - \sin^2\theta \qquad \sin^2\theta = 1 - \cos^2\theta$$

$$1 + \tan^2\theta = \sec^2\theta \qquad \tan^2\theta = \sec^2\theta - 1 \qquad 1 = \sec^2\theta - \tan^2\theta$$

$$1 + \cot^2\theta = \csc^2\theta \qquad \cot^2\theta = \csc^2\theta - 1 \qquad 1 = \csc^2\theta - \cot^2\theta$$

https://trigidentities.info

  - Pythagorean Relationships ↑
- ✓ Proving Questions

Chapter 10

Sum and difference formulas for cosine and sine

1) Cos(α±β)

Coscos Sinsin, sign different

Cos(α+β)=cosα cosβ-sinα sinβ

Cos(α-β)=cosα cosβ+sinα sinβ

i.e. Calculate cos(105˚) without using a calculator

$\cos(45+60)=\cos45 \cos60-\sin45 \sin60$

2) Sin(α±β)

Sincos Cossin, sign same

$\text{Sin}(\alpha+\beta)=\sin\alpha \cos\alpha+\cos\beta \sin\beta$

$\text{Sin}(\alpha-\beta)=\sin\alpha \cos\alpha-\cos\beta \sin\beta$

i.e. Calculate sin(4π/3) without using a calculator

$\sin(3\pi/2-\pi/6)=\sin3\pi/2 \cos3\pi/2-\cos\pi/6 \sin\pi/6$

Sum and difference formulas for tangent

1) Tan(α±β)

Sign same in numerator, different in denominator

$\text{Tan}(\alpha+\beta)= \tan\alpha+\tan\beta/1-\tan\alpha \tan\beta$

$\text{Tan}(\alpha-\beta)=\tan\alpha-\tan\beta/1+\tan\alpha \tan\beta$

i.e. evaluate tan(195˚) without using a calculator

reference angle tan(15˚)=

$\tan(60-45)=\tan60-\tan45/1+(\tan60* \tan 45)$

2)Angle between two lines: The formula Tan(α-β) can be used to find the angle between two intersecting lines. Consider the angle θ between line l_1 *and* l_2With slopes m_1 *and* m_2*. The inclination* α of l_1 *:* $\tan\alpha=m_1$ and the inclination β of l_2: $\tan\beta=m_2$. At the same time,

$\tan\theta= \tan(\alpha-\beta)= m_1- m_2/1+ m_1 m_2.$

3) Double and half-angle formulas

$\text{Sin}2\alpha=\sin(\alpha+\alpha)=2\sin\alpha\cos\alpha$

$\text{Tan}2\alpha=\tan(\alpha+\alpha)= 2\tan\alpha/1-(\tan\alpha)^2$

$\text{Cos}2\alpha=\cos(\alpha+\alpha)=\cos^2\alpha-\sin^2\alpha$

Or $2\cos^2\alpha-1$ or $1-2\sin^2\alpha$ (Both formulas got from trig identity)

$\text{Sin}\alpha/2=\pm\sqrt{}(1-\cos\alpha)/2$

$\text{Cos}\alpha/2=\pm\sqrt{}(1+\cos\alpha)/2$

$\text{Tan}\alpha/2=\pm\sqrt{}(1-\cos\alpha)/(1+\cos\alpha)$

Or $\sin\alpha/(1+\cos\alpha)$ or $(1-\cos\alpha)/\sin\alpha$

Questions you need to review for this chapter:

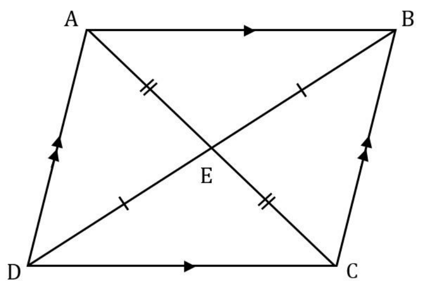
1. Trigonometric equations
   a) Use double/half angle formulas
   b) Use sum difference formula
   c) Use quadratic formula
2. Trigonometric inequality
3. Know how to prove the formulas
4. Find the supplementary angles formed by two lines

Chapter 9

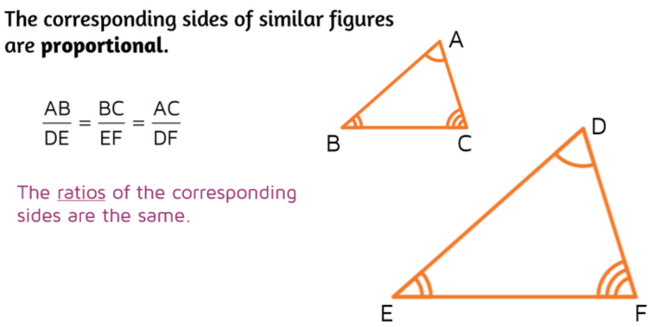
Capitalized Letter (A) is angle, Lowercase Letter (a) is the side corresponds to that angle

Solving right triangles

- Need to know how to find the length of the diagonals (measure of the angle between diagonals…) of rhombus, square, rectangle, etc.



- Similar triangles ratio
- Circumscribed circle/inscribed circle
  - c/Sin C = b/Sin B = a/Sin A = Diameter of the circumscribed circle
- Tangent lines to the circle
- Polygons
  - Central angle=360/n sides
  - Area= 1/2absinC*n sides
- Properties of an isosceles triangle
- Area of segment sector-triangle



- Area of sector(A):
  - $A=\theta/360*\pi r^2$ (degree) or $=1/2\theta r^2$(radian)

Area of Triangle= 1/2*one side*other side*sine of included angle= 1/2absinC

Law of Sine

In△ABC= sinA/a=sinB/b=sinC/c (or can be the reciprocal way:a/sinA)

-SSA situation is called the ambiguous case

How to test? (angle B given, side b is opposite to angle B, side a is adjacent)

b>asinB 2 solutins/△

b=asinB 1 solution/△

b<asinB 0 solution/△

Law of Cosine

In△ABC, to solve side

$a^2=c^2+b^2-2cb\cos A$

$b^2=a^2+c^2-2ac\cos B$

$c^2=a^2+b^2-2ab\cos C$

$(\text{side opposite angle})^2=(\text{side adjacent to angle})^2+(\text{other side adjacent to angle})^2-2(\text{one adjacent side}*\text{other adjacent side})\cos(\text{angle})$

To solve angle

$\text{Cos}C=(a^2+b^2-c^2)/2ab$

$\text{Cos }B=(a^2+c^2-c^2)/2ac$

$\text{Cos }A=(c^2+b^2-a^2)/2bc$

$\text{Cos(angle)}=[(\text{adjacent})^2+(\text{adjacent})^2-(\text{opposite})^2]/2(\text{adjacent})*(\text{adjacent})$

Given	Use	Tofind
Side, angle, side	Cos law	The third side and then one of the remaining angles
Side,side,side	Cos law	Any two angles
Angle, side, angle/ Angle, angle, side	Sin law	The remaining sides
Side, side, angle	Sin law	An angle opposite a given side and then the third side(Note that 0,1,2 triangles are possible)

Navigation and Surveying

The course of a ship or plane is the angle measured in degrees from the north direction to the direction of the ship or plane(CLOCKWISE)

The compass bearing of one location from another is measured in the same way

It is usually given as an acute angle from the north-south line toward the east-west direction.

i.e. bearing of B from A=060˚ (three digits)

bearing of A from B=240˚

Chapter 13

Arithmetic and Geometric Sequences

Sequences is a function whose domain is the set of positive integers or a subset of the positive integers that consists of the n integers 1,2,3…n

Term: each number in the sequence

i.e. a_1=the first term in the sequence

a_n=the nth term in the sequence

Arithmetic sequence:

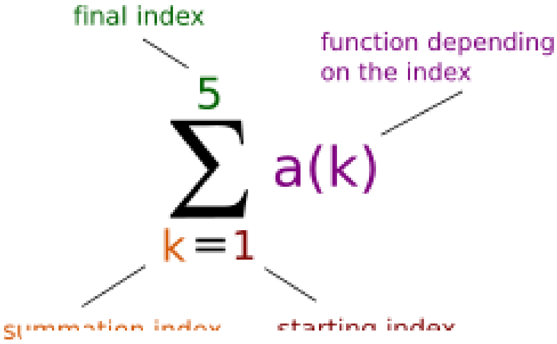
- A sequence is an arithmetic sequence if there is a constant difference(d) for which $a_n=a_{n-1}+d$ for all integers n<1
- General Formula/Explicit formula: $a_n=a_1+(n-1)d$
- Recursive formula: 1. $a_1=a_1$ an initial condition that tells where the sequence starts 2. $a_n-a_{n-1}=d$ a formula that tells how many term in the sequence is related to the proceding term
- Arithmetic mean: (a+b)/2

Geometric sequence:

- A sequence is a geometric sequence when there is a constant ratio(r) for which $a_n=a_{n-1}*r$, for all integers n>1
- General Formula/Explicit formula: $a_n=a_1*r^{n-1}$
- Recursive formula: 1. $a_1=a_1$ an initial condition that tells where the sequence starts 2. $r=a_n/a_{n-1}$ a formula that tells how many term in the sequence is related to the proceding term
- Geometric mean: √ab

Arithmetic and Geometric Series

- A series is the indicated sum of the terms of a sequence
- A series may be written using sigma Notation. The letter k is called the index of summation



Arithmetic Series Formula

$$S_n = n\left(\frac{a_1 + a_n}{2}\right)$$

$n =$ the number of terms being added

$a_1 =$ the first term

$a_n =$ the nth term (last term)

For the geometric series formula r≠1

Geometric Series Formula

$$S_n = a_1\left(\frac{1 - r^n}{1 - r}\right)$$

WHERE:

a_1 = first term

r = common ratio

n = number of terms

Limits of Infinite sequences

A sequence that does not have a last term is called infinite:

A convergent series is one where the sum approaches a finite value as more terms are added.

A divergent series is one in which the sum grows without bound or does not converge to a single value.

-If |r|<1, then $\lim_{n\to\infty} r^n = 0$

-If |r|>1 or r=±1, it is divergent with no limits

Sums of Infinite Series

The sum of an infinite series of numbers is defined to be the limit of its associated sequence of partial sums (S_n) as the number of the sequence's terms increases without bond

- For any infinite series, if the sequence of partial sums has a finite limit S, then the infinite series is said to **converge** to the sum S.
- If the sequence of partial sums approaches infinity or has no finite limit, the infinite series is said to **diverge**.

$$\sum_{k=1}^{\infty}$$

-If |r|<1, then $\lim_{n\to\infty} S^{\infty} = \frac{a}{1-r}$ for the geometric series only

-If |r|>1 or r=±1, it is divergent with no limits

Repeating decimals

Ex: Write 1.64134134134… in fraction

= 1.64+0.00 134+0.000 001 34+0.000 000 00134…

A1=0.00134   R=0.001

S=$\frac{a1}{1-r}$=$\frac{0.00134}{1-0.001}$=$\frac{0.00134}{0.999}$

$\frac{0.00134}{0.999}$+$\frac{164}{100}$=fraction for this repeating decimal

Sigma Notation

All three

Sum of natural numbers	$\sum_{n=1}^{n} n = \frac{1}{2} n(n+1)$
Sum of squares	$\sum_{n=1}^{n} n^2 = \frac{1}{6} n(n+1)(2n+1)$
Sum of cubes	$\sum_{n=1}^{n} n^3 = \frac{1}{4} n^2(n+1)^2$

Mathematical Induction

- The principle of Mathematical Induction

Let Pn be a statement involving the positive integer n. The statement is true for every n greater or equal to 1 if the following two conditions hold

1. The statement is true for n=1m that i , P1 is true
2. For any positive integer k, whenever Pk is true, then Pk+1 is true